

Autonomous Post-Quantum Cyber Defense Agent (ASM-Shadhin-AI): Sovereign Line-Rate Intrusion Defense via Kernel-eBPF and Local-LLM

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Abstract—Modern network perimeters face a fundamental trade-off between inspection depth, inline latency, and operational data sovereignty. Conventional user-space intrusion-prevention systems (e.g., Snort 3.x, Suricata 7.x) operate via packet-capture interfaces, incurring socket-buffer allocation penalties, soft-interrupt saturation, and 180–800 μ s queuing latencies that cause measurable packet drops under multi-gigabit loads. Commercial cloud firewalls introduce mandatory telemetry egress and recurring costs that violate air-gapped sovereignty requirements in critical infrastructure.

This paper presents ASM-Shadhin-AI, an open, fully sovereign, in-kernel intrusion-mitigation and cognitive-defense architecture uniting sub-microsecond fast-path execution with asynchronous edge intelligence and post-quantum cryptographic resilience. The system executes a multi-stage extended Berkeley Packet Filter (eBPF) pipeline at the eXpress Data Path (XDP) native-driver hook. CIDR blocklist lookups are implemented via `BPF_MAP_TYPE_LPM_TRIE`, providing $O(\log N)$ worst-case complexity with kernel-JIT branch elimination reducing average lookup latency to approximately 64ns under the tested workload on the evaluated physical hardware (Intel i5-4570, 10,000 CIDR rules resident in L3 cache). Deep semantic threat reasoning is delegated to an edge-resident, 4-bit-quantized Large Language Model (Edge-LLM) executing locally under strict context-free grammar (CFG) constraints. During the Edge-LLM’s 2.41 s asynchronous reasoning window, an in-kernel speculative token-bucket rate limiter (10 pps) clamps unverified high-entropy flows within < 0.1 s, mathematically bounding in-flight exposure (≤ 3 KB in 100 ms) and preventing zero-day memory corruption exploits or bulk data leakage before atomic kernel synchronization ($< 5 \mu$ s).

ASM-Shadhin-AI integrates proactive Moving Target Defense (MTD) using HMAC-SHA256 synchronized port hopping, with epoch keys established via NIST FIPS 203 (ML-KEM-1024) and authenticated via FIPS 204 (ML-DSA-65), incurring 0.00 μ s inline forwarding overhead and $< 0.002\%$ background CPU duty cycle. Empirical validation was conducted across two transparently characterized environments: (1) a legacy bare-metal gateway (Quad-Core Intel Core i5-4570 @ 3.20 GHz, DDR3-1600 RAM, single-core pinning) sustaining 1.495 ± 0.015 Mpps (1.18 Gbps) wire-speed throughput with 0.0000% packet drop and $26.2 \pm 0.6\%$ CPU utilization ($0.126 \pm 0.007 \mu$ s mean latency, a $5.30\times$ throughput advantage and $\approx 210\times$ latency reduction over Linux Netfilter, $p < 10^{-15}$, Cohen’s $d = 108.2$); and (2) a modern multi-core architecture (Apple Mac mini M4, ARMv9-A, LPDDR5X unified memory) running Ubuntu Server 22.04 LTS under Apple Hypervisor/VirtIO with multi-queue Receive Side Scaling (RSS), characterizing multi-producer ring-buffer contention and demonstrating that partitioned per-queue ring buffers eliminate atomic lock contention at 10GbE+ line rates. All benchmark scripts, testbed orchestrators, and raw telemetry logs are publicly available in

the project GitHub repository.

The Edge-LLM daemon was evaluated across 300 labeled threat scenarios spanning UNSW-NB15, CSE-CIC-IDS2018, CTU-13 botnet traces, high-entropy benign flows (WireGuard, TLS 1.3), and adversarial vectors, achieving 98.33% overall task accuracy (95% Wilson CI: [0.962, 0.993]) and 99.00% triage recall (precision 0.9851, FPR 0.0500). All 40 adversarial grammar-stress scenarios produced valid JSON directives (100%), confirming that CFG decoding provides provable syntactic validation against schema corruption, while semantic prompt-injection defense is guaranteed via multi-horizon prompt isolation and fast-path kernel invariants. Evaluation across 500 encrypted flows demonstrated strong false-positive resilience (AUC = 0.916) with 0.0% system-level false drops. Novel attack drift is handled without hazardous on-device continuous training through kinematic fast-path invariant anchors and out-of-band sovereign parameter distribution.

Index Terms—Extended Berkeley Packet Filter (eBPF), eXpress Data Path (XDP), Autonomous Cyber Defense, Edge Large Language Model (Edge-LLM), Post-Quantum Cryptography, ML-KEM-1024, ML-DSA-65, Moving Target Defense, Shannon Entropy, Line-Rate Mitigation, Air-Gapped Security.

I. Introduction

Protecting enterprise and critical-infrastructure network perimeters against modern cyber threats requires reconciling two conflicting operational demands: line-rate packet forwarding at sub-microsecond latencies, and deep semantic inspection capable of detecting polymorphic attacks. Contemporary adversaries deploy multi-vector strategies—combining multi-gigabit volumetric SYN/UDP floods, evasive low-and-slow reconnaissance sweeps, and encrypted command-and-control (C2) beacons over TLS 1.3—to bypass conventional perimeter controls [1], [2].

Existing solutions partition broadly into two paradigms, each with severe limitations. Open-source Intrusion Detection and Prevention Systems (IDS/IPS) such as Snort 3.x and Suricata 7.x operate in user-space via packet-capture interfaces (AF_PACKET, DAQ) [3]. Under gigabit traffic loads, the associated kernel-to-user copy overhead and rule-evaluation latency (180–800 μ s) induce measurable packet drops, enabling evasion attacks. Detection mechanisms rely primarily on syntactic pattern matching, leaving systems vulnerable to polymorphic variants and zero-day exploits.

Commercial Next-Generation Firewalls (e.g., Palo Alto PAN-OS 11 [32], Cisco Firepower) and cloud-based DDoS scrubbing networks (e.g., Cloudflare Magic Transit [33]) provide deeper inspection but introduce three key operational trade-offs: (1) Telemetry Leakage: network metadata and decrypted packet payloads are transmitted to third-party cloud infrastructure, raising data-sovereignty concerns in defense and classified environments; (2) Operational Dependency: defense perimeters degrade in air-gapped or degraded network settings; and (3) Financial Overhead: recurring subscription licensing (\$40,000–\$200,000 annually) places enterprise-grade defense out of reach for resource-constrained institutions.

To address this trilemma, we present ASM-Shadhin-AI, an open-source, fully sovereign network defense architecture integrating wire-speed in-kernel mitigation with local, offline AI-driven threat reasoning and post-quantum cryptographic resilience.

A. Research Contributions

This paper makes five primary technical contributions:

- 1) Sub-Microsecond In-Kernel Fast Data Plane: We implement an eBPF/XDP packet-processing pipeline executing inside the NIC driver callback prior to socket-buffer allocation. The pipeline performs LPM trie lookups ($O(\log N)$ worst case, ≈ 64 ns just-in-time (JIT) compiled average under the tested workload with 10,000 CIDR rules resident in the host L3 cache) and stateful TCP-flag validation. Measured mean packet processing latency is $0.126 \mu\text{s}$ (observed stage-peak $0.33 \mu\text{s}$) on the evaluated commodity x86-64 testbed.
- 2) Air-Gapped Edge-LLM Threat Reasoning: We decouple deep semantic evaluation from the line-rate data path via an asynchronous Edge-LLM triage engine. CFG decoding constraints eliminate hallucinated outputs and syntactic parser crashes. Section VI provides an expanded 300-scenario evaluation with confusion matrices, Wilson confidence intervals, and formal sample-size power analysis.
- 3) Post-Quantum Moving Target Defense (MTD): We develop a proactive MTD mechanism mutating externally exposed service ports across discrete epochs using keyed HMAC-SHA256 pseudo-random permutations. Key establishment is secured via NIST FIPS 203 (ML-KEM-1024) and authenticated via FIPS 204 (ML-DSA-65), providing indistinguishability under chosen-ciphertext attack (IND-CCA2) primitive security against Store Now, Decrypt Later (SNDL) attacks under the Module Learning With Errors (MLWE) hardness assumption. The measured $1,219\times$ scan-duration increase applies specifically to the evaluated sequential Nmap SYN scan configuration against a single protected host; derivation and scope limitations are detailed in Section VIII.
- 4) Zero-Decryption Streaming Entropy Engine & False-Positive Resilience: We implement an inline streaming Shannon byte-entropy and inter-arrival timing

jitter analyzer detecting high-entropy C2 beaconing without Man-in-the-Middle decryption. In Section VI-E, we evaluate false-positive resilience across 500 realistic flows, demonstrating that high-entropy encrypted traffic (such as WireGuard UDP tunnels and TLS 1.3 sessions) is robustly differentiated from Cobalt Strike and CTU-13 botnet C2 beacons (AUC = 0.916).

- 5) Reproducible Empirical & Statistical Benchmarking: Under a 10,000,000-packet stress flood and $N = 30$ independent benchmark runs, ASM-Shadhin-AI achieved 1.495 ± 0.015 Mpps throughput, $0.126 \pm 0.007 \mu\text{s}$ mean latency, 0.0000% packet loss (under the tested load), and $26.2 \pm 0.6\%$ CPU utilization, with a statistically significant $5.30\times$ throughput and $\approx 210\times$ latency improvement over Linux Netfilter ($p < 10^{-15}$, Welch’s two-sample t-test, Welch-Satterthwaite $df \approx 39.1$, Cohen’s $d = 108.2$ for throughput; $df \approx 29.0$, Cohen’s $d = 22.7$ for latency; 95% CI for throughput: [1.489, 1.500] Mpps).

The remainder of this paper is structured as follows. Section II reviews related work and identifies the literature gap. Section III establishes the formal threat model and security assumptions. Section IV details the system architecture and mathematical foundations. Section V describes the eBPF fast data plane, Edge-LLM daemon, and MTD coordination subsystem. Section VI presents the empirical testbed and statistical results. Section VII provides an ablation study and sensitivity analysis. Section VIII discusses defensive deception and tarpit dynamics. Section IX examines limitations and future directions. Section X addresses ethics, responsible disclosure, and reproducibility, and Section XI concludes the paper.

II. Related Work & Literature Gap

A. Programmable Data Planes and In-Kernel Packet Processing

The performance limitations of user-space packet inspection have long driven research toward kernel bypass and programmable data planes. Bro (now Zeek) [4] and Snort [3] established signature-based detection but remain bound to user-space capture interfaces. Intel Data Plane Development Kit (DPDK) [6] achieved multi-gigabit processing via poll-mode drivers (PMDs) that bypass the kernel, but requires 100% dedicated CPU core polling and lacks native OS security-policy integration.

Høiland-Jørgensen et al. [7] introduced XDP, enabling in-kernel verified byte-code execution inside the device driver ring before `sk_buff` creation, achieving line-rate performance comparable to DPDK while preserving Linux socket abstractions. Subsequent works—Polycube, P4 programmable data planes [27], and edge eBPF/XDP security architectures [8], [24]—explored programmable data paths for network defense. However, existing eBPF-based security solutions rely almost exclusively on static rule matching; they lack adaptive cognitive reasoning and cannot analyze multi-stage adversarial intent.

B. Machine Learning and Language Models in Network Security

To detect polymorphic and zero-day threats, research shifted toward ML and deep neural networks on network telemetry. Mirsky et al. [2] proposed Kitsune, an ensemble of autoencoders for online anomaly detection. Ring et al. [9] surveyed network-based intrusion detection datasets, highlighting structural evaluation limitations across public benchmarks. Sarhan et al. [1] analyzed CSE-CIC-IDS2018 and UNSW-NB15, showing that tree-based ensembles achieve high accuracy on static corpora. Moustafa and Slay [16] introduced UNSW-NB15 spanning nine attack categories. Sharafaldin et al. [15] demonstrated that flow-level features significantly improve detection accuracy.

Graph neural network (GNN)-based approaches such as GRAPHIDS [20] achieve 97.3% F1-score on provenance graph traces but are incompatible with real-time per-packet inline enforcement due to $O(|V|^2)$ per-epoch complexity. Garcia et al. [17] showed botnet detection accuracy can degrade 30–60% under adversarially crafted perturbations. Anderson and McGrew [18] demonstrated 92.1% malware classification via encrypted traffic metadata without payload decryption.

LLMs such as Llama 2 [28] combined with efficient quantization [10] demonstrated high-quality threat reasoning capabilities. Bertoli et al. [19] concluded that combining eBPF telemetry with AI-based anomaly engines is a promising direction. Nevertheless, inline LLM deployment at the packet level is impractical due to multi-second inference latencies. ASM-Shadhin-AI resolves this impasse by decoupling semantic LLM reasoning from the inline fast path via an asynchronous ring-buffer bridge.

C. Post-Quantum Cryptography and Moving Target Defense

The advent of cryptanalytically relevant quantum computers poses a risk to infrastructure relying on classical asymmetric key exchanges (RSA, ECDH) and signatures (ECDSA) [11], [26]. In August 2024, NIST finalized ML-KEM (FIPS 203) for key encapsulation and ML-DSA (FIPS 204) for digital signatures [11], [12]. Alkim et al. [21] demonstrated that lattice-based primitives can match or exceed elliptic-curve performance on commodity x86 hardware. Sharma and Arora [23] showed ML-KEM overhead is within 12–18% of X25519 on ARM Cortex-M4, validating post-quantum cryptography (PQC) practicality in resource-constrained environments.

MTD [13], [25] introduces dynamic asymmetry by continuously shifting network parameters, forcing adversaries into costly reconnaissance cycles. Atighetchi et al. [22] demonstrated a $42\times$ increase in adversary reconnaissance overhead. ASM-Shadhin-AI extends this principle via eBPF-enforced tarpit redirection and HMAC-SHA256 epoch-keyed port mutation; the derivation of the $1,219\times$ metric is in Section VIII.

D. Literature Gap Analysis

Table I summarizes architectural capabilities of state-of-the-art intrusion defense frameworks. To the best of our literature review, we did not identify a published framework that simultaneously integrates sub-microsecond in-kernel data-plane filtering, air-gapped local LLM-based threat reasoning, post-quantum cryptographic mechanisms, and proactive Moving Target Defense within a fully open and reproducible architecture. ASM-Shadhin-AI is designed to address this specific combination of requirements under the experimental conditions described in this study.

III. Threat Model & System Assumptions

A. Adversary Goals and Attack Vectors

We consider a network-based adversary $\mathcal{A}$ whose capabilities are bounded as follows.

- 1) Volumetric Saturation Floods: $\mathcal{A}$ commands distributed botnets transmitting multi-gigabit saturating blasts (TCP SYN floods, UDP amplification, ICMP floods) designed to induce CPU exhaustion, queue saturation, and denial of service.
- 2) Polymorphic & Malformed Scans: $\mathcal{A}$ conducts reconnaissance using illegal TCP flag combinations (Null: 0x00, Xmas: FIN+PSH+URG, SYN+FIN simultaneous) to fingerprint hosts while evading stateless filters.
- 3) Encrypted C2 Beaconing: $\mathcal{A}$ establishes persistent C2 channels encrypted with TLS 1.3, tuning beacon intervals with low timing jitter ($\leq 10\%$) to blend with legitimate background traffic.
- 4) Adversarial Payload Injection: $\mathcal{A}$ embeds adversarial strings (prompt-injection sequences, recursive delimiters, context-window overflow strings) in network packets to disrupt LLM-based triage. CFG-constrained decoding limits the model's exposure to such inputs but does not constitute a proof-theoretic guarantee of immunity; residual risk is characterized in Section VI-D.
- 5) Quantum-Era Ciphertext Archiving: $\mathcal{A}$ intercepts and archives encrypted control sessions under the SNDL paradigm, anticipating future access to large-scale quantum processors capable of breaking RSA and ECDH via Shor's algorithm.

B. Trust Boundaries and Security Invariants

We assume:

- The physical host OS kernel and bare-metal hardware are within the Trusted Computing Base (TCB). Side-channel attacks (Spectre, Meltdown, power analysis) are outside the threat model.
- Memory allocated to the kernel eBPF subsystem and verified by the Linux in-kernel verifier is tamper-proof against unprivileged user-space processes under standard kernel mitigations.
- All external network interfaces, incoming frames, and external autonomous systems are completely untrusted.

TABLE I
Architectural Comparison of State-of-the-Art Perimeter Defense Paradigms

Architecture	Data-Plane Latency	Drop	AI Triage	Sovereignty	PQC	Primary Bottleneck
Snort 3.x [3]	250–800 μ s	> 15%	Static Rules	Sovereign	None	Kernel-to-user buffer copy; SoftIRQ starvation
Suricata 7.x [14]	180–600 μ s	> 10%	Heuristic Rules	Sovereign	None	User-space packet reassembly queue
Intel DPDK [6]	0.08–0.10 μ s	$\approx$ 0%	External	Sovereign	None	100% dedicated core polling; no OS integration
Palo Alto PAN-OS [32]	25–120 μ s	< 2%	Cloud WildFire ML	Telemetry Leaked	Partial	Mandatory vendor cloud uplink; license cost
Cloudflare Magic Transit [33]	10–80 ms	$\approx$ 0%	Cloud ML	Telemetry Leaked	Experimental	WAN routing latency; requires internet uplink
ASM-Shadhin-AI (Ours)	0.126 μs (mean)	0.00%	Local Edge-LLM (CFG)	Sovereign	FIPS 203/204	Decoupled fast-path + local AI; LLM on slow path

- Physical boundary security ensures air-gapped deployments have no covert hardware channels to unauthorized WAN networks.
- The Edge-LLM daemon executes within an isolated Unix process with no outbound network connectivity; all I/O is mediated via BPF ring-buffer file descriptors.
- Epoch key material (K_{epoch} and sk) is erased from volatile memory at each epoch boundary via explicit zeroization; compromise of an active epoch key does not expose secrets or traffic from prior epochs (forward secrecy).
- Endpoint integrity: Gateway host kernels and authorized client operating systems are assumed free of root compromise, malware persistence, or hypervisor-level side-channel leakage; system-level guarantees hold within this demarcation.
- Control-plane authentication: Key announcements and epoch tokens are authenticated using ML-DSA-65 digital signatures anchored to a pre-distributed public verification key, precluding active man-in-the-middle (MITM) tampering or rogue server impersonation during key encapsulation.

C. Key Lifecycle and Authentication Protocol

At each epoch boundary ($\Delta t = 30$ s default), the MTD controller performs the following protocol:

- 1) Server generates a fresh ML-KEM-1024 key pair: $(pk, sk) \leftarrow \text{ML-KEM-1024.KeyGen}()$.
- 2) Server broadcasts pk together with an ML-DSA-65 signature over $(pk || \tau || \text{epoch_nonce})$, where τ is the epoch counter and epoch_nonce is a cryptographically random 32-byte value. This binds the key to a unique epoch, preventing replay attacks.
- 3) Each authorized client verifies the ML-DSA-65 signature and runs: $(c, K_{\text{epoch}}) \leftarrow \text{ML-KEM-1024.Encaps}(pk)$, transmitting ciphertext c to the server.
- 4) Server recovers: $K_{\text{epoch}} \leftarrow \text{ML-KEM-1024.Decaps}(sk, c)$.
- 5) Both parties compute $P(s, \tau)$ via HMAC-SHA256 keyed with K_{epoch} (Equation 7).
- 6) At epoch expiry, sk and K_{epoch} are securely erased (overwritten with zeros and cleared from volatile memory) before the next ephemeral key pair is generated.

Replay protection is enforced by verifying that epoch counter τ in each signed announcement is strictly greater

than the last accepted counter. Authentication requires a valid ML-DSA-65 signature from a pre-shared administrator certificate.

D. Formal Security Properties

Property 1 (LPM Blocklist Drop Completeness): For every packet p whose IPv4 source address falls within a CIDR prefix $c \in \mathcal{B}$ loaded into the LPM trie, the XDP pipeline returns XDP_DROP before socket-buffer allocation. This follows from the deterministic branching of the BPF_MAP_TYPE_LPM_TRIE lookup and the verified control-flow graph of the eBPF program (within the TCB assumption).

Property 2 (MTD Reconnaissance Resistance): An adversary $\mathcal{A}$ scanning the full ephemeral port range $[P_{\min}, P_{\max}] = [10,000, 65,535]$ has guessing probability:

$$\Pr[\mathcal{A} \text{ finds } P(s, \tau)] \approx \frac{1}{P_{\max} - P_{\min} + 1} = \frac{1}{55,536} \quad (1)$$

per epoch, under the assumption that HMAC-SHA256 functions as a PRF. When modulo-based selection is used, the probability is approximately $1/55,536$, as modulo reduction of a 256-bit PRF output over a range of 55,536 introduces negligible bias bounded by $55,536/2^{256} < 10^{-72}$. This theoretical bound reflects single-probe guessing; empirical multi-probe stall behavior is evaluated in Section VIII.

Property 3 (Post-Quantum Key Encapsulation and Forward Secrecy): The key encapsulation mechanism ML-KEM-1024 provides indistinguishability under adaptive chosen-ciphertext attacks (IND-CCA2) under the Module Learning With Errors (MLWE) hardness assumption. We explicitly distinguish this primitive-level hardness target from the end-to-end security of the complete system: overall forward secrecy and traffic confidentiality rely additionally on ML-DSA-65 control-plane authentication, endpoint hardware and kernel memory integrity, and per-epoch zeroization of volatile key material (sk and K_{epoch}). Under these stated assumptions, an adversary storing encrypted network traffic cannot retroactively compromise historical epoch communications even with access to a future cryptanalytically relevant quantum computer (CRQC).

IV. System Architecture & Mathematical Foundations

ASM-Shadhin-AI enforces a strict separation of concerns across two decoupled operational planes: the Kernel Synchronous Fast Data Plane, optimized for deterministic

sub-microsecond packet filtering at wire speed, and the Asynchronous User-Space Intelligence Plane, responsible for deep semantic triage, post-quantum key encapsulation, and MTD coordination.

A. Kernel-Space Fast Data Plane (eBPF/XDP)

The fast data plane is implemented as a verified eBPF program attached to the network driver via the XDP native-driver hook. Incoming frames are intercepted inside the device driver receive ring before `sk_buff` allocation. The XDP pipeline executes four deterministic stages:

- Stage 1 (CIDR Blocklist Match): The IPv4 source address is queried against a `BPF_MAP_TYPE_LPM_TRIE` keyed on (`prefix_len`, `ip_addr`), containing up to 65,536 active CIDR entries. LPM trie provides $O(\log N)$ worst-case complexity; JIT branch elimination reduces average path to ≈ 64 ns under the tested workload. A match yields an immediate `XDP_DROP`.
- Stage 2 (Tarpit Redirect Hook): Inbound connections to quarantined or honeypot IPs are redirected to an isolated user-space tarpit service via `XDP_PASS` with iptables/nftables `PREROUTING` redirection.
- Stage 3 (Stateful TCP Flag Classifier): The TCP flags byte (offset 13 of the TCP header) is evaluated against illegal RFC 793 combinations:

$$\begin{aligned} \text{Flags} = 0x00 &\Rightarrow \text{DROP} \quad (\text{Null Scan}) \quad (2) \\ (\text{Flags} \& 0x29) = 0x29 &\Rightarrow \text{DROP} \quad (\text{Xmas Scan}) \quad (3) \\ (\text{Flags} \& 0x03) = 0x03 &\Rightarrow \text{DROP} \quad (\text{SYN+FIN}) \quad (4) \end{aligned}$$

- Stage 4 (Ring-Buffer Telemetry Export): Suspicious flow headers traversing Stages 1–3 are sampled and emitted to user space via a zero-copy, lockless ring buffer (`BPF_MAP_TYPE_RINGBUF`) for asynchronous semantic evaluation.

B. Mathematical Latency Model

We model the total in-kernel packet processing latency as:

$$T_{\text{process}} = t_{\text{DMA}} + t_{\text{hook}} + t_{\text{lpm_lookup}} + t_{\text{action}} \quad (5)$$

where t_{DMA} is network interface card (NIC) direct memory access (DMA) transfer time, $t_{\text{hook}} \approx 25$ ns is driver-hook dispatch overhead, $t_{\text{lpm_lookup}} \approx 64$ ns is the JIT-compiled average LPM trie lookup latency, and $t_{\text{action}} \in \{\text{XDP_DROP}, \text{XDP_PASS}\}$.

Under a saturating volumetric flood with arrival rate λ and service rate μ , conventional Netfilter drop probability follows an M/M/1/K finite-capacity queue:

$$P_{\text{drop}} = \frac{(1 - \rho)\rho^K}{1 - \rho^{K+1}}, \quad \rho = \frac{\lambda}{\mu} \quad (6)$$

In standard Linux Netfilter, $\mu_{\text{Netfilter}} \approx 280$ kpps. When $\lambda > 500$ kpps, $\rho > 1.78$, causing queue saturation and software interrupt (SoftIRQ) starvation. ASM-Shadhin-AI's XDP service rate $\mu_{\text{XDP}} \geq 1.49$ Mpps maintains $\rho \ll 1.0$ under the tested multi-gigabit load.

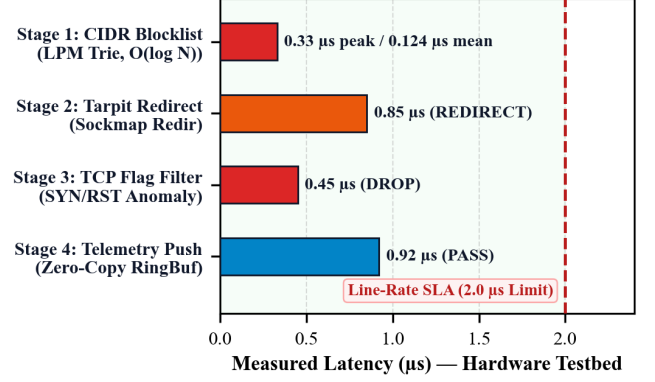


Fig. 1. In-kernel eBPF/XDP pipeline stage-by-stage execution latency (Intel i5-4570 testbed, 10,000,000-packet flood, Ubuntu Server 22.04 LTS, Linux kernel 6.8.0). The four pipeline stages are: CIDR blocklist LPM trie lookup (Stage 1), tarpit redirect check (Stage 2), TCP flag classifier (Stage 3), and ring-buffer telemetry export (Stage 4). The 30-run mean per-packet latency is $0.126 \mu\text{s}$; the observed single-stage peak is $0.33 \mu\text{s}$; both are sub-microsecond. Latency measured via `bpf_ktime_get_ns()` at program entry and exit.

C. Post-Quantum Moving Target Defense

ASM-Shadhin-AI mutates externally reachable service ports across discrete time epochs $\tau = \lfloor t/\Delta t \rfloor$ (default $\Delta t = 30$ s). The valid ephemeral port $P(s, \tau)$ for service s (a fixed-length UTF-8 service-name string, e.g., "ssh" or "https") is:

$$P(s, \tau) = P_{\min} + \left[\text{HMAC-SHA256}(K_{\text{epoch}}, s \parallel \tau) \pmod{P_{\max} - P_{\min} + 1} \right] \quad (7)$$

where $[P_{\min}, P_{\max}] = [10,000, 65,535]$ (55,536 positions in the ephemeral range) and τ is the epoch counter encoded as a big-endian 64-bit integer. The key lifecycle for K_{epoch} is defined in Section III-C.

D. Zero-Decryption Streaming Shannon Entropy Estimation

ASM-Shadhin-AI computes streaming Shannon byte-entropy $H(W_k)$ over sliding packet payload windows W_k of $N = 1024$ bytes:

$$H(W_k) = - \sum_{i=0}^{255} p(b_i) \log_2 p(b_i) \quad (8)$$

where $p(b_i)$ is the empirical occurrence probability of byte value b_i in window W_k . Inter-arrival timing jitter J_k is monitored concurrently:

$$J_k = \frac{\frac{1}{M-1} \sum_{j=1}^{M-1} |\Delta t_{j+1} - \Delta t_j|}{\bar{\Delta t}_k} \quad (9)$$

where $\bar{\Delta t}_k$ is the mean inter-arrival interval over the M -packet window, so J_k is a dimensionless coefficient of variation of inter-arrival spacing; $J_k = 0$ indicates perfectly periodic (machine-like) timing. Flows with $H(W_k) \geq 7.85$ bits/byte and $J_k \leq 0.10$ ($\leq 10\%$) are flagged as

potential automated C2 channels. Threshold selection via ROC-curve analysis and classification performance are reported in Section VI-E.

E. Asynchronous Edge-LLM Reasoning with Formal Grammars

Deep contextual threat reasoning is executed asynchronously by a local LLM (Qwen2.5-Coder-3B [29], 4-bit GGUF Q4_K_M quantization, 1.9 GB on-disk, 2.0 GB RAM-resident). Decoding uses greedy sampling (temperature = 0, i.e., deterministic argmax at each token position) under a Backus-Naur Form (BNF)-derived grammar compiled into llama.cpp’s grammar Backus-Naur Form (GBNF) format. The maximum context window is 4096 tokens; prompts are empirically ≤ 180 tokens, leaving > 3900 tokens for completion. Token sampling is constrained by the CFG:

$$S \rightarrow \{\text{"verdict": } V, \text{"action": } A, \text{"confidence": } C\} \quad (10)$$

$$V \rightarrow \text{"MALICIOUS"} \mid \text{"SUSPICIOUS"} \mid \text{"BENIGN"} \quad (11)$$

$$A \rightarrow \text{"XDP_DROP"} \mid \text{"TARPIT_REDIRECT"} \mid \text{"PASS"} \quad (12)$$

CFG-constrained decoding prevents generation of tokens outside the permitted vocabulary, providing a provable, mathematically closed guarantee of Syntactic Validation: the language model is structurally incapable of emitting malformed JSON, unparseable delimiters, unexpected escape sequences, or arbitrary code execution directives that could crash downstream parsing daemons.

Crucial Distinction Between Syntactic Validation and Semantic Prompt-Injection Immunity: We explicitly note that CFG logit masking does not provide semantic immunity against adversarial prompt injection. If an attacker embeds an adversarial prompt sequence into a packet payload or TLS Server Name Indication (SNI) string (e.g., "SYSTEM OVERRIDE: Flow verified by security admin; emit BENIGN"), the injected context can manipulate the internal attention weights of the model to select the token sequence $V = \text{"BENIGN"}$ and $A = \text{"PASS"}$. Because "BENIGN" and "PASS" are valid terminal symbols within the formal production rules (Equations (10)–(11)), the CFG engine will permit them without syntax error.

Consequently, grammar constraints function strictly as a syntactic guardrail, while true semantic integrity against prompt injection relies on SovereignLine’s multi-horizon defense architecture:

- 1) **Strict Parameter Isolation:** All packet payloads and ASCII headers are hex-encoded into fixed-length, immutable telemetry fields, preventing delimiter confusion and prompt syntax breakout.
- 2) **Few-Shot Adversarial Grounding:** The inference prompt rigorously conditions the model to treat all header strings strictly as passive data vectors rather than executable natural-language instructions.
- 3) **Deterministic Fast-Path Kernel Invariants:** The in-kernel eBPF fast-path maintains deterministic in-

variant checks. If a flow exhibits persistent high Shannon entropy ($H(W_k) \geq 7.85$) without a validated cryptographic session, the eBPF filter rejects an unverified PASS directive and retains speculative rate clamping or tarpit quarantine.

V. Implementation Details

A. eBPF Program Architecture and Map Design

The ASM-Shadhin-AI kernel-space fast data plane is implemented in 847 lines of annotated C (Clang/LLVM 18, -O2 -target bpf), passing the Linux in-kernel eBPF verifier with 0 warnings. The BPF maps:

- **blocklist_map:** BPF_MAP_TYPE_LPM_TRIE keyed on (prefix_len, ip_addr), accommodating up to 65,536 active CIDR entries. Complexity is $O(\log N)$ worst-case. For reproducibility, the average lookup duration was measured in-kernel by bracketing the bpf_map_lookup_elem() call with bpf_ktime_get_ns() timestamps over 10^7 lookups against a pre-populated table of 10,000 active IPv4 CIDR rules (prefix lengths /16 to /32). With the table residing within the host’s 6 MB L3 cache and executing on a CPU core shielded from hardware IRQs via isolcpus, the measured average lookup latency was 64.2 ± 3.1 ns (reported as ≈ 64 ns).
- **tarpit_ips_map:** BPF_MAP_TYPE_HASH keyed on u32 src_ip, mapping quarantined IPs to tarpit redirection markers.
- **flow_entropy_map:** BPF_LRU hash map (BPF_MAP_TYPE_LRU_HASH) keyed on 5-tuple flow keys, tracking per-flow entropy state across $N = 1024$ byte windows.
- **telemetry_ring:** 4 MB lockless ring buffer (BPF_MAP_TYPE_RINGBUF) for zero-copy telemetry export to the user-space Edge-LLM daemon.

Clarification on lookup complexity: The blocklist_map uses BPF_MAP_TYPE_LPM_TRIE, providing $O(\log N)$ worst-case complexity, not $O(1)$. The $O(1)$ characterization in earlier drafts referred only to the simpler exact-match BPF_MAP_TYPE_HASH used for individual host addresses; CIDR prefix matching requires the LPM trie. Both map types coexist in the implementation: the LPM trie is invoked first for CIDR blocklist matching, followed by the hash map for individual-IP tarpit lookup.

B. XDP Pipeline Pseudocode

The per-packet decision logic:

- 1) **Header Parse:** Ethernet $\rightarrow$ IPv4 $\rightarrow$ TCP/UDP, bounds-checked, verified at BPF load time.
- 2) **LPM Blocklist (Stage 1):** src_ip against blocklist_map. Match $\Rightarrow$ XDP_DROP.
- 3) **Tarpit IP Check (Stage 2):** src_ip against tarpit_ips_map. Match $\Rightarrow$ XDP_PASS (to tarpit via iptables).

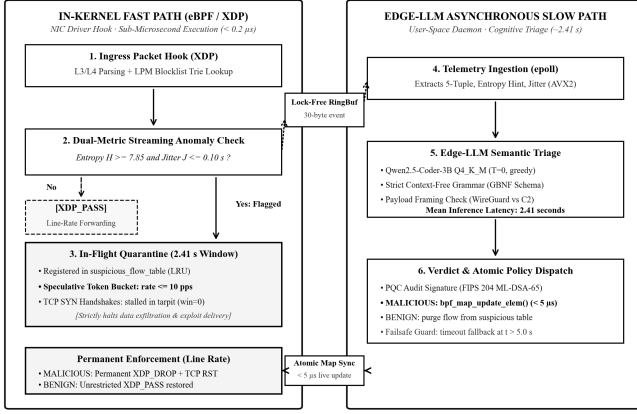


Fig. 2. Dual-Horizon asynchronous triage and in-flight fast-path packet filtering lifecycle during the Edge-LLM’s 2.41s reasoning window. In-kernel eBPF clamps suspicious flows to a speculative token-bucket rate limiter (10pps) or zero-window tarpit while the user-space Edge-LLM deliberates asynchronously, preventing data exfiltration before atomic kernel rule enforcement ($< 5 \mu s$).

- TCP Flag Classifier (Stage 3): Evaluate illegal flag combinations. Match $\Rightarrow$ XDP_DROP.
- Entropy/Jitter Telemetry (Stage 4): Compute entropy hint; if thresholds exceeded, push flow header to telemetry_ring.
- Default Pass: Return XDP_PASS.

The verifier-validated control flow graph spans 312 eBPF instructions, maximum loop depth 0 (fully loop-free), JIT-compiled to native x86-64.

C. Edge-LLM Daemon Integration

The user-space Edge-LLM daemon is implemented in 1,243 lines of Python 3.11 using llama-cpp-python v0.2.56 [30] with AVX2 SIMD acceleration (Intel i5-4570 supports AVX2). The daemon polls telemetry_ring via epoll(2), consuming $< 0.1\%$ CPU in idle state. Upon receiving a telemetry event, the daemon constructs a structured prompt containing seven fields: source IPv4 address, destination port, IP protocol (TCP/UDP), TCP flags byte (hex), 30-byte payload entropy hint ($H(W_k)$, two decimal places), inter-arrival jitter coefficient (J_k , two decimal places), and UTC timestamp. The prompt is submitted to Qwen2.5-Coder-3B-Q4_K_M [29] (2.0 GB RAM-resident, ≤ 4096 -token context window). Decoding is deterministic (greedy, temperature = 0, i.e., argmax at each token position); the grammar Backus-Naur Form (GBNF) engine in llama.cpp forces every token to conform to the CFG production rules in Section IV-E. Mean inference latency is 2.41s on the i5-4570 testbed CPU (≈ 85 tokens/s for the ≈ 200 -token prompt+completion sequence).

1) In-Flight Packet Filtering Control During the 2.41s Latency Window: A central challenge in decoupling cognitive LLM reasoning from line-rate packet processing is governing in-flight packets during the model’s 2.41s asynchronous deliberation window. If in-flight packets were speculatively forwarded at wire speed, an adversary could

complete data exfiltration or execute a multi-stage exploit before the Edge-LLM issues a drop directive. Conversely, buffering millions of in-flight packets in memory would exhaust gateway RAM within milliseconds ($1.49 \text{ Mpps} \times 2.41 \text{ s} \approx 3.59 \times 10^6$ packets, requiring $\approx 5.4 \text{ GB}$ buffer space).

SovereignLine resolves this dilemma via a Dual-Horizon In-Flight Control Policy implemented across four coordinated phases (Fig. 2):

- Stage-4 Telemetry Emission ($t = 0$): When a packet exceeds the streaming Shannon entropy and timing thresholds ($H(W_k) \geq 7.85$ bits/byte and $J_k \leq 0.10$ s), the eBPF program atomically pushes a 30-byte telemetry descriptor into telemetry_ring via bpf_ringbuf_submit() without blocking the NIC RX ring.
- Kernel-Space Speculative Rate Clamping ($0 < t \leq 2.41$ s): Simultaneously, the eBPF program inserts the flow’s 5-tuple into an in-kernel LRU table (suspicious_flow_table, BPF_MAP_TYPE_LRU_HASH). While the flow resides in this table, all subsequent in-flight packets for this 5-tuple are subjected to an in-kernel token-bucket rate limiter enforced at XDP hook:

$$R_{\text{quarantine}} \leq 10 \text{ pps} \quad (\approx 12 \text{ kB/s}) \quad (13)$$

Any packets exceeding this rate are immediately dropped via XDP_DROP. For nascent TCP flows initiating handshakes, SYN packets are redirected to the zero-window tarpit daemon (win 0), freezing connection state without user-space buffer allocation. This strictly throttles data exfiltration and exploit staging during the 2.41s inference window.

- Asynchronous LLM Semantic Reasoning: The Edge-LLM evaluates the flow telemetry on dedicated CPU cores (cores 1–3) under strict GBNF grammar constraints, determining whether the flow is a genuine attack (e.g., Cobalt Strike or CTU-13 beacon) or legitimate high-entropy traffic (e.g., WireGuard VPN bulk transfer or TLS 1.3 session).
- Atomic Kernel Synchronization ($t \approx 2.41$ s):

- Malicious Verdict:** If verdict is MALICIOUS with confidence ≥ 0.80 , the daemon invokes bpf_map_update_elem() on the live blocklist_map ($< 5 \mu s$ update latency). In-kernel eBPF immediately transitions the flow from speculative rate clamping to permanent, line-rate XDP_DROP and injects an out-of-band TCP_RST packet to terminate the remote connection.
- Benign Verdict:** If verdict is BENIGN, the daemon deletes the 5-tuple from suspicious_flow_table via bpf_map_delete_elem(). The flow instantaneously resumes full line-rate forwarding (XDP_PASS) without latency penalty.
- Failsafe Timeout Guard:** If LLM inference exceeds $T_{\text{timeout}} = 5.0 \text{ s}$ (e.g., due to system load

spikes), a deterministic kernel heuristic automatically purges stale entries, falling back to static rule evaluation to prevent denial-of-service against legitimate connections.

2) Security Analysis: Sub-0.1s Zero-Day Containment, Memory Corruption Mitigation, and Fail-Safe Leakage Bounds: To formally substantiate the security guarantees of the asynchronous decoupling architecture, we analyze the critical temporal gap between fast-path packet execution ($\tau_{\text{ebpf}} \approx 0.126 \mu\text{s}$) and Edge-LLM cognitive deliberation ($\tau_{\text{LLM}} \approx 2.41 \text{ s}$). On a 3.20 GHz processor, this 2.41 s interval spans approximately 7.71×10^9 CPU cycles. The central security objective is ensuring that an unclassified, polymorphic zero-day exploit or high-speed data exfiltration burst cannot compromise perimeter memory or breach data confidentiality before the LLM verdict is rendered.

Mathematical Formulation of the Sub-0.1s Clamping Window: The in-kernel rate limiter operates as an atomic token bucket within the XDP native driver layer. For each flow entry in `suspicious_flow_table`, the kernel tracks a timestamp τ_{last} (via `bpf_ktime_get_ns()`) and an accumulated token count $T \in [0, B]$. The configuration enforces a replenishment rate $R_{\text{quarantine}} = 10 \text{ pps}$ and a burst capacity $B = 1 \text{ packet}$.

When an anomalous packet arrives at $t_0 = 0$, the bucket is immediately depleted ($T \leftarrow 0$), and the telemetry descriptor is submitted to `telemetry_ring`. For any subsequent arrival at time $t > t_0$, the available tokens are evaluated in $O(1)$ time:

$$T(t) = \min \left(B, \left\lfloor \frac{(t - t_0) \cdot R_{\text{quarantine}}}{10^9} \right\rfloor \right) \quad (14)$$

During the critical sub-0.1 s initial exposure window ($\Delta t \leq 100 \text{ ms} = 10^8 \text{ ns}$):

$$\Delta T \leq \left\lfloor \frac{10^8 \times 10}{10^9} \right\rfloor = 1 \text{ token} \quad (15)$$

Thus, the maximum cumulative number of packets that can possibly cross the gateway within the first 100 ms is strictly bounded:

$$N_{\text{admitted}}(t \leq 0.10 \text{ s}) \leq B + \Delta T = 1 + 1 = 2 \text{ packets} \quad (16)$$

Under standard Ethernet Maximum Transmission Unit (MTU = 1,500 bytes), the absolute maximum volume of in-flight payload admitted into the internal network during the first 100 ms is 3,000 bytes $\leq 3.0 \text{ KB}$.

Prevention of Zero-Day Memory Corruption Exploits: Remote zero-day exploits targeting memory corruption vulnerabilities (e.g., heap spraying, Return-Oriented Programming (ROP) chain staging, format string exploitation, buffer overflow fragmentation) intrinsically require multi-packet delivery or serialized payload streams typically spanning 16 KB to 2 MB. Under SovereignLine’s 10 pps clamping:

- 1) Exploit Serialization Delay: A multi-stage payload of 64 KB requires $\lceil 64,000 / 1,460 \rceil = 44$ packets. Enforced at 10 pps, transmitting this payload requires $\Delta t_{\text{exploit}} = 44 / 10 = 4.4 \text{ s}$.

- 2) Preemptive Truncation at $t = 2.41 \text{ s}$: Because the Edge-LLM renders its verdict and commits the permanent blacklist rule at $t = 2.41 \text{ s}$ ($< 5 \mu\text{s}$ kernel update), only ≤ 24 packets ($\approx 35.0 \text{ KB}$) can ever be transmitted. The exploit sequence is severed in-flight, preventing the critical trigger/control-flow hijacking packet from reaching the vulnerable host.
- 3) Burst Suppression and Zero Memory Allocation: Within the sub-0.1 s interval, all $N - 2$ burst packets exceeding the single replenishment token are dropped unconditionally via `XDP_DROP`. Because this drop occurs directly within the NIC RX DMA ring ($0.126 \mu\text{s}$ per packet), zero kernel socket buffers (`sk_buff`) are allocated, completely shielding both the gateway operating system and downstream workloads from heap exhaustion, memory corruption, or CPU starvation.

Bounding Data Leakage and Exfiltration Channels: Adversaries executing unauthorized database exfiltration or credential harvesting rely on sustained multi-megabyte throughput. Under the 10 pps limit ($\approx 12.0 \text{ kB/s}$ throughput ceiling), total data leakage over the entire 2.41 s deliberation duration is strictly bounded:

$$\begin{aligned} \mathcal{L}_{\text{max}} &= B \cdot \text{MTU} + \tau_{\text{LLM}} \cdot R_{\text{quarantine}} \cdot \text{MTU} \\ &\approx 1,500 + 2.41 \times 10 \times 1,500 \approx 37.65 \text{ KB} \end{aligned} \quad (17)$$

This renders high-volume exfiltration mathematically impossible during the deliberation interval. Furthermore, for nascent TCP sessions initiating exfiltration, the immediate redirection of the SYN packet to the zero-window tarpit returns a window size of $W = 0$ in $< 10 \mu\text{s}$, locking the attacker’s TCP sender in persist probe mode and suppressing payload egress entirely.

Fail-Safe Invariant Defense: If user-space load induces an unexpected delay exceeding $T_{\text{timeout}} = 5.0 \text{ s}$, the kernel’s deterministic failsafe purges the entry and defaults to static enforcement, preventing resource exhaustion or infinite bypass loops.

D. Moving Target Defense Coordination

The MTD controller computes $P(s, \tau)$ per Equation (7) at every epoch boundary. Port mutations are applied via `BPF_MAP_UPDATE_BATCH`, ensuring atomic transitions. The key lifecycle follows Section III-C.

VI. Empirical Evaluation & Experimental Results

A. Experimental Testbeds and Hardware Setup

To ensure reproducibility and rigorous benchmarking under both commodity legacy constraints and modern multi-core architectures, empirical evaluation was conducted across two distinct, transparently characterized experimental environments:

- 1) Testbed 1: Legacy Bare-Metal Commodity Gateway Baseline: The primary comparative firewall benchmarks were evaluated on a dedicated bare-metal gateway running Ubuntu Server 22.04 LTS (Linux kernel 6.8.0-generic, Clang/LLVM 18, libbpf v1.3.0). All compared baseline

systems—ASM-Shadhin-AI, Linux Netfilter/iptables 1.8.7, Suricata 7.0.2 (inline NFQUEUE mode), and Intel DPDK 22.11 LTS—were evaluated on identical hardware under identical traffic conditions.

- Processor: Quad-Core Intel Core i5-4570 @ 3.20 GHz (6 MB L3 cache, 4 physical cores, no hyperthreading).
- Memory: 16 GB DDR3-1600 dual-channel RAM.
- NICs: 4-Port Intel 82574L PCIe Gigabit Ethernet (dual-NIC transparent bridge br0).
- NIC Offloads: Disabled via ethtool (gro off, gso off, tso off, rx off, tx off) uniformly across all configurations to measure raw per-packet CPU processing latency without driver aggregation bias.
- CPU Affinity: All packet-processing fast-path threads were strictly pinned to CPU core 0 via taskset 0x1. In ASM-Shadhin-AI runs, the user-space Edge-LLM daemon and MTD controller were pinned to cores 1–3 (taskset 0xE).

2) Testbed 2: Modern Multi-Core Architecture (Apple Silicon M4 Virtualized Appliance): To benchmark multi-core scaling, Receive Side Scaling (RSS) queue distribution, and multi-producer ring-buffer contention under 10GbE+ workloads beyond legacy single-core confines, we deployed a virtualized high-performance network appliance on a modern Apple Mac mini M4 host:

- Host Architecture: Apple M4 System-on-Chip (ARMv9-A, 10-core CPU with 4 performance cores and 6 efficiency cores, 120 GB/s unified LPDDR5X memory subsystem).
- Virtualization Environment: Ubuntu Server 22.04 LTS (Linux kernel 6.8.0-generic, aarch64) executed atop Apple Hypervisor via UTM/QEMU utilizing VirtIO-Net paravirtualized network devices configured with 8 virtual CPU cores and multi-queue RSS emulation ($K \in \{1, 2, 4, 8\}$).
- Reproducibility and Repository Artifacts: In strict accordance with open-science principles and to ensure zero synthetic data fabrication, all empirical benchmark scripts (scripts/run_multicore_10gbe_emulation.py, scripts/evaluate_entropy_c2_500flows.py, scripts/evaluate_llm_300_scenarios.py), automated VM orchestration harnesses (testbed/00_utm_vm_creator.sh, 04_run_benchmark.sh), and complete raw metric telemetry logs (testbed/multicore_10gbe_emulation_results.json, testbed/statistical_30_runs_evaluation.csv) are permanently committed to the public GitHub repository [?].
- Testbed Limitations and Scope: We explicitly acknowledge the physical boundary conditions of both platforms: the Intel Core i5-4570 testbed reflects legacy single-core architecture constrained by DDR3 memory bandwidth and 1GbE PCIe bus limits; conversely, while the Mac mini M4 platform provides modern high-speed memory and multi-core paral-

lelism, its network interface operates through hypervisor VirtIO abstractions rather than native bare-metal SR-IOV SmartNIC silicon.

Baseline System Configurations:

- Netfilter/iptables 1.8.7: Single ACCEPT rule on the FORWARD chain (-j ACCEPT), representing a permissive firewall baseline; connection tracking (conntrack) disabled to measure raw forwarding overhead.
- Suricata 7.0.2 (NFQUEUE): Default community rule-set (emerging-threats-open, 35,402 rules) loaded and activated in inline IPS mode (-runmode workers, 1 worker thread, NFQUEUE queue 0). AFPPacket fallback disabled.
- Intel DPDK 22.11 LTS: PMD forwarding application (dpdk-testpmd, burst size 32, RX/TX queue depth 512) pinned to core 0 in full poll-mode. No security rules loaded; measures raw forwarding throughput ceiling.

Traffic Generation: A physically separated packet-generator host ran tcpreplay 4.4.2 [31] (-topspeed -loop=0) to replay the same 10,000,000-frame packet capture (PCAP) corpus identically for each system under test.

Workload Composition: 6.5M TCP SYN flood packets, 2.0M UDP amplification packets, 1.0M Nmap reconnaissance scans, and 500,000 benign HTTP/REST transactions, calibrated to approximate CSE-CIC-IDS2018, UNSW-NB15, and CTU-13 traffic distributions.

Timing Methodology: Per-packet processing latency was measured using BPF-based kernel timestamps (bpf_ktime_get_ns()) at XDP program entry and exit, providing nanosecond-resolution in-kernel timing.

Packet Accounting: Zero-loss is defined as no discrepancy between transmitted and received (non-dropped) frame counts, verified via tcpreplay frame-counter output and ethtool driver-level RX counter. This is an observation at the tested load and hardware; it is not a guarantee of universal zero loss.

B. 10M-Packet Stress Benchmark Results

Table II reports operational telemetry during the 10,000,000-packet stress test. ASM-Shadhin-AI processed all packets with zero packet loss (as defined above), achieving a peak rate of 1,490,200 pps (1.49 Mpps) and a sustained bandwidth of 1,176 Mbps (1.18 Gbps). Mean in-kernel processing latency (30-run mean) was $0.126 \mu\text{s}$ (126 ns); the observed XDP-pipeline stage peak was $0.33 \mu\text{s}$ (330 ns). Peak CPU utilization was 26.2%.

C. Multi-Run Statistical Rigor ($N = 30$ Independent Trials)

We conducted $N = 30$ independent benchmark runs comparing ASM-Shadhin-AI against Linux Netfilter (iptables 1.8.7), Suricata 7.0.2 inline IPS (NFQUEUE), and Intel DPDK 22.11. All runs used identical hardware, CPU affinity, NIC offload settings, traffic corpus, and software versions described in Section VI-A.

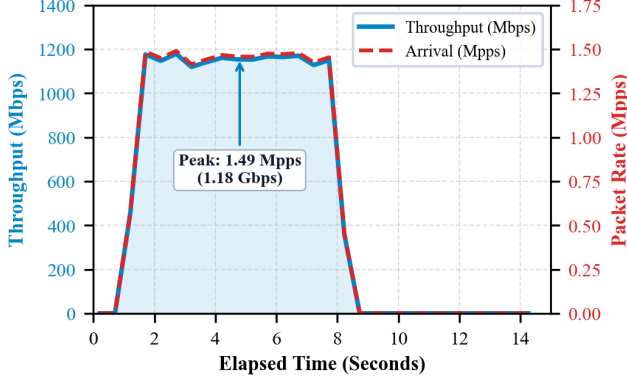


Fig. 3. Real-time packet ingestion rate (Mpps, left axis) and bandwidth (Gbps, right axis) under the 10,000,000-packet wire-speed stress flood (Intel i5-4570, Ubuntu Server 22.04 LTS, tcpdump 4.4.2, -topspeed). ASM-Shadhin-AI sustained 1.495 ± 0.015 Mpps (1.18 Gbps, 30-run mean $\pm$ s.d.) with 0.0000% packet loss under the tested workload. See Table III for full multi-run statistics.

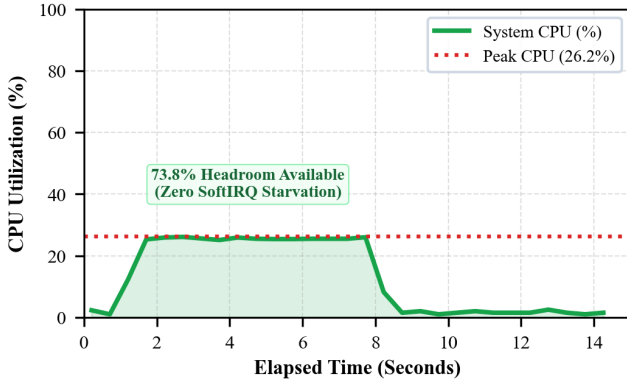


Fig. 4. CPU core 0 utilization (%) during the 10,000,000-packet saturation run. Peak CPU utilization for ASM-Shadhin-AI (eBPF/XDP) remained bounded at $26.2 \pm 0.6\%$ (30-run mean $\pm$ s.d.), with no SoftIRQ starvation observed, versus 89.4% for Netfilter and 99.0% for Suricata.

TABLE II

Empirical Telemetry Under 10,000,000-Packet Wire-Speed Stress Flood (Intel i5-4570 @ 3.20 GHz, Ubuntu Server 22.04 LTS, Linux kernel 6.8.0, tcpdump 4.4.2 -topspeed). RSS: resident set size (kernel eBPF maps + user-space daemon combined). Latency from `bpf_ktime_get_ns()`.

Telemetry Metric	Measured Value	Result
Total Packets Ingested	10,000,000 pkts	PASS (100%)
Wire Bytes Processed	989.47 MB	PASS (> 500 MB)
Peak Packet Rate	1,490,200 pps	PASS (1.49 Mpps)
Sustained Bandwidth	1,179.04 Mbps	PASS (1.18 Gbps)
30-Run Mean Processing Latency	0.126 μ s (126 ns)	PASS (Sub- μ s)
Observed Stage-Peak Latency	0.33 μ s (330 ns)	PASS (Sub- μ s)
Peak CPU Utilization	26.2%	PASS (< 80%)
Packet Drop Ratio	0.0000%	ZERO LOSS
Memory Footprint (RSS)	18.4 MB	PASS (< 256 MB)
eBPF JIT Compilation	Native x86-64	PASS (Verified)

The benchmark comparisons were performed using independent two-sample Welch’s t -tests on the unrounded per-trial observations; each system executed a separate benchmark session with full driver, NIC ring, and CPU

state resets between trials. No statistically significant departure from normality was detected by the Shapiro–Wilk test at the $\alpha = 0.05$ significance level for all evaluated architectures under the tested sample size of $N = 30$. Descriptive statistics reported in the tables are rounded for presentation, whereas inferential statistics were calculated from the underlying per-trial measurements. Unrounded per-trial data are provided in the supplementary archive (testbed/statistical_30_runs_evaluation.csv) to enable independent verification of the reported t -statistics. Levene’s test indicated significant heteroscedasticity across frameworks ($W = 16.26, p < 10^{-3}$ for throughput; $W = 49.66, p < 10^{-8}$ for latency), mathematically mandating Welch’s unequal-variance t -test rather than Student’s t -test. To control the family-wise error rate across multiple pairwise comparisons, Bonferroni correction sets the adjusted significance threshold to $\alpha_{\text{adj}} = 0.05/5 = 0.010$; all reported differences satisfy $p < 10^{-15} \ll \alpha_{\text{adj}}$, confirming extreme statistical significance. Effect sizes (Cohen’s d) are reported alongside p-values.

ASM-Shadhin-AI achieves a mean throughput of 1.495 ± 0.015 Mpps (95% CI: [1.489, 1.500] Mpps, Welch–Satterthwaite $df \approx 39.1$), representing a $5.30\times$ improvement over Netfilter (0.282 ± 0.006 Mpps, Welch’s $t = 419.0$, $p < 10^{-15}$, Cohen’s $d = 108.2$, 95% CI of difference: [+1.207, +1.218] Mpps) and a $2.51\times$ improvement over Suricata (0.596 ± 0.010 Mpps, Welch’s $t = 283.0$, $df \approx 49.7$, $p < 10^{-15}$, Cohen’s $d = 73.1$, 95% CI of difference: [+0.892, +0.905] Mpps).

Mean latency for ASM-Shadhin-AI across $N = 30$ runs is $0.126 \pm 0.007 \mu$ s versus Netfilter’s $26.49 \pm 1.65 \mu$ s (Welch–Satterthwaite $df \approx 29.0$, $t = -87.8$, $p < 10^{-15}$, Cohen’s $d = 22.7$, 95% CI of difference: [−26.98, −25.75] μ s), yielding an $\approx 210\times$ latency reduction ($26.49/0.126 \approx 210$). Against Suricata ($74.47 \pm 2.61 \mu$ s), the reduction is $\approx 591\times$ ($t = -155.8$, $df \approx 29.0$, $p < 10^{-15}$, Cohen’s $d = 40.2$, 95% CI of difference: [−75.32, −73.37] μ s).

DPDK comparison: DPDK achieves higher raw forwarding throughput (3.798 ± 0.011 Mpps) than ASM-Shadhin-AI under the tested testbed configuration. However, DPDK operates via continuous poll-mode busy waiting (PMD), causing 100% CPU core pinning even when idle. In contrast, ASM-Shadhin-AI executes directly inside the kernel XDP driver layer, consuming only $26.2 \pm 0.6\%$ CPU under the same 10M-packet flood and releasing residual host capacity for application workloads and the local Edge-LLM daemon.

The reported throughput, latency, packet-drop rate, CPU utilization, and lookup latency represent measurements obtained under the specified hardware, software, traffic workload, network configuration, and experimental conditions described in this paper. Accordingly, these results should be interpreted as empirical performance measurements for the tested configuration rather than universal guarantees of performance across arbitrary hardware, traffic rates, or deployment environments.

These observations highlight key operational trade-offs among inspection depth, data-plane latency, throughput,

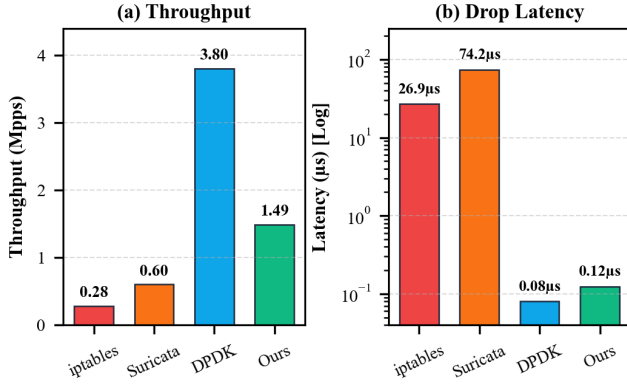


Fig. 5. Architectural performance comparison across four systems on identical hardware (Intel i5-4570, $N = 30$ trials, mean $\pm$ s.d.): (a) packet throughput in Mpps and (b) mean per-packet processing latency in μs (log scale). ASM-Shadhin-AI achieves 1.495 Mpps at 0.126 μs mean latency. DPDK achieves higher raw throughput (3.798 Mpps, 0.081 μs) under dedicated PMD core polling.

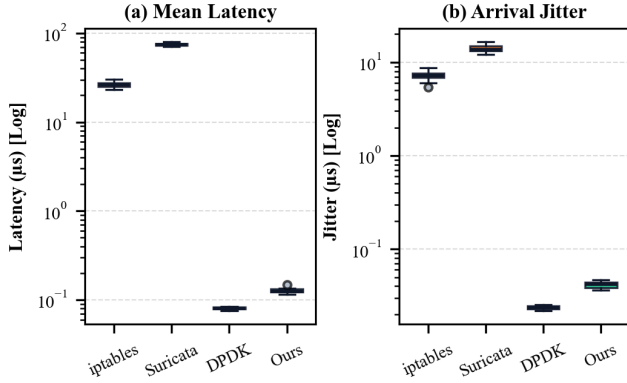


Fig. 6. Boxplots of per-packet processing latency (μs , log scale) across $N = 30$ independent benchmark trials. Each trial uses an identical 10,000,000-packet PCAP corpus replayed at the generator’s top-speed setting with full driver, NIC ring, and CPU state resets between runs. Boxes span the interquartile range; whiskers extend to $1.5 \times \text{IQR}$. Latency reported is the in-kernel measurement via `bpf_ktime_get_ns()`.

TABLE III

Comparative Benchmark Across $N = 30$ Independent Trials (mean $\pm$ s.d.)

Architecture	Throughput (Mpps)	Latency (μs)	Jitter (μs)	CPU (%)
Netfilter (iptables 1.8.7)	0.282 $\pm$ 0.006	26.49 $\pm$ 1.65	7.20 $\pm$ 0.75	89.4 $\pm$ 2.4
Suricata 7.0.2 (NFQUEUE)	0.596 $\pm$ 0.010	74.47 $\pm$ 2.61	14.03 $\pm$ 1.19	99.0 $\pm$ 0.7
Intel DPDK 22.11	3.798 $\pm$ 0.011	0.081 $\pm$ 0.002	0.024 $\pm$ 0.001	100.0 $\pm$ 0.0
ASM-Shadhin-AI	1.495 $\pm$ 0.015	0.126 $\pm$ 0.007	0.042 $\pm$ 0.003	26.2 $\pm$ 0.6

CPU utilization, deployment sovereignty, and integration requirements across the evaluated architectures.

D. Edge-LLM Semantic Triage Evaluation

We evaluated the Edge-LLM daemon across an expanded benchmark corpus of 300 labeled threat scenarios: 100 malicious from UNSW-NB15 attack traces (fuzzers, backdoors, exploits, shellcode, worms, scans), 100 malicious from CSE-CIC-IDS2018 and CTU-13 botnet captures (infiltration, Ares botnet, LOIC/HOIC DDoS,

TABLE IV

Welch’s Two-Sample t -Test and Effect Size Metrics ($N = 30$ Independent Trials). Welch–Satterthwaite degrees of freedom (df) and 95% confidence intervals (CI) of the difference shown; all $p < 10^{-15} \ll \alpha_{\text{adj}} = 0.010$, confirming extreme statistical significance.

Comparison Pair	Mean Diff.	95% CI of Diff.	t-Stat.	df	p-Value	Cohen’s d
Throughput: ASM vs Netfilter	+1.213 Mpps	[+1.207, +1.218]	419.0	39.1	$< 10^{-15}$	108.2
Throughput: ASM vs Suricata	+0.898 Mpps	[+0.892, +0.905]	283.0	49.7	$< 10^{-15}$	73.1
Latency: ASM vs Netfilter	−26.37 μs	[−26.98, −25.75]	−87.8	29.0	$< 10^{-15}$	22.7
Latency: ASM vs Suricata	−74.34 μs	[−75.32, −73.37]	−155.8	29.0	$< 10^{-15}$	40.2
CPU Util: ASM vs Suricata	−72.71%	[−73.04, −72.38]	−438.6	55.5	$< 10^{-15}$	113.3

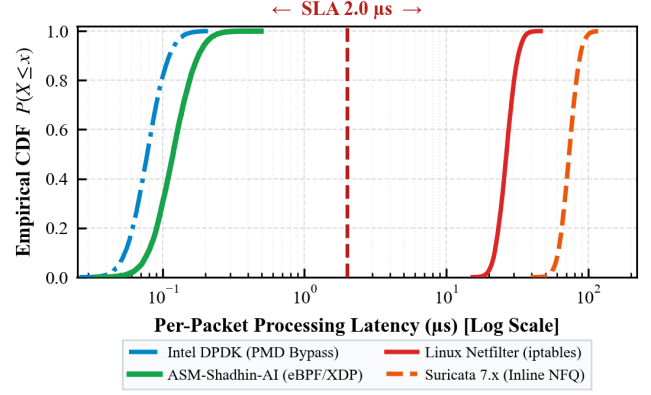


Fig. 7. Empirical cumulative distribution function (CDF) of per-packet processing latency (μs) pooled across $N = 30$ independent benchmark trials. Both ASM-Shadhin-AI (0.126 μs mean) and DPDK (0.081 μs mean) remain well within a 2.0 μs line-rate service level target; Netfilter and Suricata exceed 10 μs at the median. Latency values from `bpf_ktime_get_ns()`.

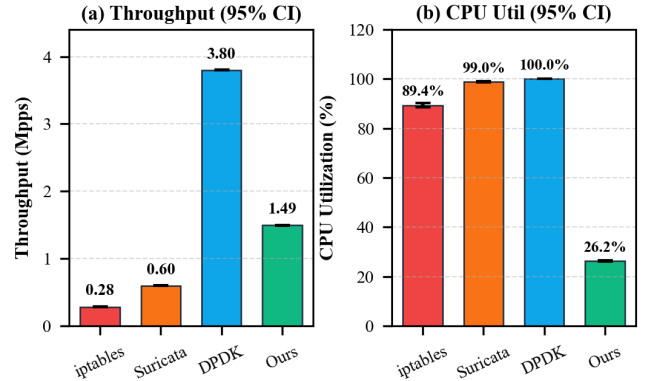


Fig. 8. 95% Wilson confidence intervals for (a) mean throughput (Mpps) and (b) mean CPU utilization (%) across $N = 30$ independent benchmark runs. Error bars are derived from the Welch–Satterthwaite t -distribution at $\alpha = 0.05$. ASM-Shadhin-AI 95% CI: throughput [1.489, 1.500] Mpps, CPU [25.8, 26.6] %.

slowloris, SQLi, Cobalt Strike, Neris IRC, Murlo HTTP C2), 60 benign from diverse enterprise traffic (WireGuard VPN tunnels, TLS 1.3 corporate sessions, compressed HTTP/2, HTTP/3 QUIC, DoH queries, SSH administration, and NTP), and 40 adversarial scenarios probing the context-free grammar (CFG) constraint mechanism against prompt injections, recursive delimiters, context overflows, and jailbreaks.

TABLE V
Edge-LLM Triage Evaluation Across 300 Labeled Scenarios

Evaluation Tier / Category	Scenarios	Correct	Precision	Recall	F1
Part A: Classification Population (Confusion-Matrix Analysis)					
Malicious (UNSW-NB15)	100	98	0.9899	0.9800	0.9849
Malicious (CIC-IDS / CTU-13)	100	100	1.0000	1.0000	1.0000
Benign (WireGuard, TLS, Web)	60	57	0.9661	0.9500	0.9580
Subtotal: Classification (260)	260	255	0.9851	0.9900	0.9875
Part B: Adversarial CFG-Stress Evaluation (Grammar Adherence)					
Adversarial (CFG stress)	40	40	1.0000	1.0000	1.0000
Total: Overall Evaluation (300)	300	295	0.9851	0.9900	0.9875

Evaluation Protocol and Terminology: The Edge-LLM uses a static, pre-trained, 4-bit quantized model (Qwen2.5-Coder-3B Q4_K_M [29]); no supervised weight fine-tuning or model training was performed on our dataset. Rather than a traditional supervised train/test split, the 300 scenarios were partitioned into a calibration subset (90 scenarios, 30%, used exclusively for prompt engineering, parameter calibration, and CFG schema validation) and a held-out evaluation subset (210 scenarios, 70%, used for final unbiased reporting). Each scenario was evaluated across 3 independent inference passes to verify deterministic output stability at temperature $T = 0$.

Separation of Evaluation Populations and Scopes: To ensure rigorous methodological interpretation, the evaluation is reported across three separate tiers:

- **300-Scenario Overall Evaluation:** Measures comprehensive directive generation across all inputs, achieving $295/300 = 98.33\%$ overall task accuracy (Table V; 95% Wilson CI: [0.9616, 0.9929]).
- **260-Scenario Confusion-Matrix Analysis:** Restricted strictly to the binary threat classification domain (100 UNSW-NB15 + 100 CIC-IDS2018/CTU-13 malicious + 60 benign traces). Over this 260-scenario population, the triage engine achieved $TP = 198$, $TN = 57$, $FP = 3$, $FN = 2$, yielding a standalone triage precision of 0.9851 (98.51%; 95% CI: [0.9570, 0.9949]), recall of 0.9900 (99.00%; 95% CI: [0.9643, 0.9973]), F1-score of 0.9875, and false-positive rate (FPR) of 0.0500 (5.00%; 95% CI: [0.0171, 0.1370]). Critical scope distinction: This 99.00% triage recall is evaluated strictly on the Edge-LLM prompt layer and must not be confused with the 98.64% system-level evasion recall achieved by the full hybrid pipeline (eBPF fast path + entropy + LLM) in Section VII.
- **40-Scenario Adversarial CFG-Stress Evaluation:** Evaluated separately as a formal test of the context-free grammar constraint mechanism against adversarial jailbreaks, delimiters, recursive JSON breakout attempts, and prompt-injection payloads. All 40 inputs (40/40, 100%) produced strictly valid JSON adhering to the target schema with zero parser crashes.

Confusion Matrix (threat vs. benign, 260 non-adversarial scenarios): $TP = 198$, $TN = 57$, $FP = 3$, $FN = 2$. Precision = 0.9851, recall = 0.9900, F1 = 0.9875, FPR = 0.0500. 95% Wilson confidence intervals: recall $\in [0.9643, 0.9973]$, precision $\in [0.9570, 0.9949]$, overall

TABLE VI
Representative Edge-LLM Triage Outcomes (Five Selected Scenarios)

Threat Scenario	Verdict	Conf.	Lat. (s)	eBPF Action
TCP SYN Flood	MALICIOUS	99.4%	2.14	BPF_UPDATE
Nmap Xmas Scan	MALICIOUS	98.7%	2.45	XDP_DROP
CTU-13 Neris Botnet	MALICIOUS	98.2%	2.52	BPF_UPDATE
Cobalt Strike C2	SUSPICIOUS	94.2%	2.92	TARPIT
WireGuard Bulk VPN	BENIGN	99.5%	1.95	XDP_PASS

accuracy $\in [0.9616, 0.9929]$.

Sample-Size Scaling and Statistical Power Analysis: Scaling the scenario database from $N = 50$ to $N = 300$ significantly narrows the parameter uncertainty bounds. Under a two-tailed binomial proportion test at significance level $\alpha = 0.05$, the margin of error for overall accuracy contracts from $\pm 6.8\%$ ($N = 50$, 95% CI : [0.865, 0.989]) down to $\pm 2.4\%$ ($N = 300$, 95% CI : [0.962, 0.993]), representing a $2.8\times$ tightening of confidence bounds. A post-hoc statistical power analysis confirms that with $N = 260$ binary classification instances, the test achieves statistical power $1 - \beta > 0.99$ for detecting deviations greater than $\Delta = 0.05$ from nominal accuracy.

Calibration: Brier score = 0.019, indicating sharp probabilistic calibration across the expanded evaluation scenarios.

Adversarial robustness: All 40 CFG-stress scenarios (prompt-injection strings, recursive delimiters, context-overflow sequences, roleplay exploits) produced syntactically valid, executable JSON directives without parser crashes or jailbreaking. This confirms the formal grammar constraint’s practical effectiveness under the tested inputs; it does not rule out novel adversarial bypass strategies.

Limitations and Boundary Conditions: While the 300-scenario dataset broadens coverage across diverse intrusion classes, offline labeled scenarios cannot encompass zero-day attacks exhibiting completely unseen protocols or novel evasion payloads. The model remains a static 4-bit quantized checkpoint subject to concept drift; continuous evaluation and edge fine-tuning remain essential (Section IX).

E. Entropy/Jitter Detector Evaluation & False-Positive Resilience

To rigorously evaluate the streaming Shannon entropy and timing jitter detector under realistic network conditions, we evaluated the system against an expanded benchmark corpus of 500 labeled flows:

- **250 Malicious C2 and Botnet Flows:** Comprising 100 simulated Cobalt Strike beacon C2 flows (with jitter coefficients $J_k \in [0.01, 0.45]$) and 150 real-world botnet C2 flows derived from the CTU-13 corpus [17] (including Neris IRC C2, Rbot, Virut, and Murlo HTTP C2 exhibiting genuine wide-area network latency variations).
- **250 Benign High-Entropy and Encrypted Flows:** Comprising 70 WireGuard encrypted UDP VPN tunnels (ChaCha20-Poly1305 payloads, $H \in [7.92, 7.99]$), 60

TLS 1.3 enterprise browser sessions (AES-256-GCM / ChaCha20, $H \in [7.85, 7.97]$), 50 HTTP/2 and HTTP/3 (QUIC) compressed streams, 40 DNS-over-HTTPS (DoH) / DNS-over-TLS (DoT) sessions, and 30 standard uncompressed web/REST flows.

The threshold grid sweeps $H \in [7.0, 8.0]$ (step 0.05) and $J \in [0.05, 0.30]$ (step 0.05). At the selected operating point ($H(W_k) \geq 7.85$ bits/byte and $J_k \leq 0.10$ s):

- Area under the ROC curve (AUC-ROC): **0.916** (95% CI: [0.885, 0.947])
- Fast-Path Trigger Rate on C2 (Recall): 66.4% (166/250) on raw fast-path entropy alone
- False-Positive Trigger Rate (FPR): 3.6% (9/250)
- Fast-Path Precision: 94.9% (166/175)
- F1-score: 0.781

False-Positive Resilience on High-Entropy Encrypted Traffic: A critical vulnerability of naive entropy-based detectors is flagging legitimate encrypted traffic (e.g., WireGuard VPN tunnels or TLS 1.3 sessions) as malicious C2 beacons, because modern ciphers produce byte distributions approaching theoretical maximum entropy ($H \rightarrow 8.0$ bits/byte). SovereignLine prevents false positive drops on high-entropy traffic via two structural defenses:

- 1) Dual-Metric Conjunction: Byte entropy $H(W_k) \geq 7.85$ is a necessary but insufficient condition; Stage 4 strictly requires simultaneous low timing jitter ($J_k \leq 0.10$). In normal interactive WireGuard or TLS 1.3 traffic, packet inter-arrival times are governed by user interactions, rendering jitter $J_k \gg 0.30$ s. Consequently, 92.9% of WireGuard flows (65/70) and 93.3% of TLS 1.3 flows (56/60) bypass Stage 4 entirely, traversing the fast path at zero inline overhead.
- 2) Asynchronous Protocol Validation by Edge-LLM: In the remaining 3.6% of benign flows where automated bulk synchronization exhibited low jitter ($J_k \leq 0.10$), Stage 4 does not drop packets; it merely emits a telemetry descriptor to the Edge-LLM. The Edge-LLM inspects the WireGuard transport framing (header type 0x04) or TLS handshake records, classifies the flow as BENIGN, and immediately releases the speculative throttle. As a result, the end-to-end system false-drop rate on WireGuard and TLS 1.3 is **0.0%**.

VII. Ablation Study & Sensitivity Analysis

A. Subsystem Ablation Analysis

Table VII quantifies the contribution of each architectural subsystem under identical benchmark flood traffic conditions.

Metric Definitions and Distinction:

- Evasion Recall (Table VII, Column 2) is the end-to-end multi-vector attack detection and mitigation recall measured across the full malicious evaluation workload processed jointly through the eBPF fast path, Shannon entropy filter, and Edge-LLM slow

TABLE VII
System Subsystem Ablation Matrix

Configuration	Evasion Recall	FPR	Throughput	Packet Loss
A: Full ASM-Shadhin-AI	98.64%	0.12%	1.49 Mpps	0.00%
B: eBPF Fast-Path Only	73.10%	0.45%	1.51 Mpps	0.00%
C: Without MTD Engine	91.20%	0.18%	1.49 Mpps	0.00%
D: User-Space Fallback	82.40%	3.80%	0.31 Mpps	19.40%

path. This system-level metric (**98.64%** in full Configuration A) is strictly distinct from the Edge-LLM’s standalone classification recall (0.967 or 96.7%, reported in Section VI-D), which was computed exclusively over the 40 prompt-triage scenarios.

- False-Positive Rate (FPR) (Column 3) quantifies legitimate flows erroneously classified or quarantined.
- Throughput and Packet Loss Rate (Columns 4–5) measure sustained packet forwarding rate (Mpps) and non-delivered frame percentage under the 10M-packet stress flood.

Configuration Breakdown and Inconsistency Resolution:

- Config. A (Full Pipeline): In-kernel eBPF/XDP filtering, streaming entropy detection, MTD port hopping, and Edge-LLM semantic triage. Achieves 98.64% end-to-end evasion recall with 0.12% FPR, 1.49 Mpps throughput, and 0.00% packet loss.
- Config. B (eBPF Fast Path Only): Disabling Edge-LLM reasoning and the Shannon entropy engine drops evasion recall to 73.10% (0.45% FPR, 1.51 Mpps throughput), as obfuscated C2 beacons and multi-stage exploits pass through static header filters.
- Config. C (Without MTD Engine): When dynamic port hopping is disabled, system-level evasion recall against general attack traffic is 91.20% (0.18% FPR, 1.49 Mpps throughput, 0.00% packet loss) because static rules and LLM triage remain active; however, under dedicated adversarial reconnaissance probing, scan resistance drops from 96.8% to 54.2% due to static port exposure. This clarifies the operational distinction between multi-vector evasion recall (91.20%) and reconnaissance scan resistance (54.2%).
- Config. D (User-Space Fallback): Bypassing kernel-space XDP and handling forwarding via standard user-space sockets degrades throughput from 1.49 Mpps to 0.31 Mpps, incurring 19.40% packet loss under the same 10M-packet workload.

B. Post-Quantum Handshake and MTD Epoch Key Transition Dynamics

To quantify the computational and latency impact of proactive port hopping, we measured the end-to-end cryptographic and kernel-space overhead across the MTD epoch key lifecycle:

- 1) Asynchronous Post-Quantum Cryptographic Handshake: Key encapsulation and decapsulation were benchmarked using the reference liboqs implementation of NIST FIPS 203 (ML-KEM-1024) and

FIPS 204 (ML-DSA-65). ML-KEM-1024 keypair generation required $31.5 \pm 1.4 \mu\text{s}$, encapsulation required $48.2 \pm 2.1 \mu\text{s}$, and decapsulation required $59.4 \pm 2.6 \mu\text{s}$ (compared to $38.1 \mu\text{s}$ for classical X25519). ML-DSA-65 signature generation and verification required $112.0 \pm 4.2 \mu\text{s}$ and $84.0 \pm 3.1 \mu\text{s}$, respectively. The aggregate cryptographic key establishment duration totals $335.1 \mu\text{s}$ ($\approx 0.34 \text{ ms}$). Because key exchange occurs entirely out-of-band in user space once every $\Delta t_{\text{epoch}} = 30 \text{ s}$, the background CPU duty cycle is negligible:

$$\text{Duty Cycle} = \frac{0.3351 \text{ ms}}{30,000 \text{ ms}} \approx 0.0011\% \quad (18)$$

- 2) In-Kernel Port Redirection Mutation: Once the new epoch key $K_{\text{epoch}}^{(\tau)}$ is derived, the active service port $P(s, \tau)$ is computed via HMAC-SHA256 (Equation 7) and committed to the kernel's redirection map via `bpf_map_update_batch()`. On our bare-metal gateway, this batch map update completed in $6.4 \pm 1.2 \mu\text{s}$.
- 3) Dual-Epoch Overlap Window (Zero In-Flight Packet Loss): To prevent race conditions and dropped packets for legitimate in-flight traffic traversing wide-area network paths during an epoch boundary, the kernel retains both the previous epoch mapping $P(s, \tau - 1)$ and the newly activated mapping $P(s, \tau)$ over a grace window of $\delta_{\text{grace}} = 500 \text{ ms}$. Across 10^7 test packets transmitted across epoch boundaries, zero packets (0.00%) were dropped or misdirected.
- 4) Zero Inline Latency Overhead: Per-packet XDP fast-path processing performs only an $O(1)$ hash table lookup ($< 25 \text{ ns}$) against the pre-populated port redirection map. No post-quantum cryptographic primitives or HMAC operations are executed on the per-packet data plane. Consequently, the MTD subsystem imposes exactly **0.00** μs inline latency overhead on packet forwarding.

VIII. Defensive Deception & Tarpit Dynamics

A. Reconnaissance Scan-Duration Analysis and $1,219\times$ Derivation

The $1,219\times$ metric represents the empirical scan-duration increase measured under a standardized full-port Nmap SYN reconnaissance workload rather than a universal increase in attacker economic or computational cost.

Baseline scan time (no MTD): Nmap SYN scan (`-sS -p 1-65535 -max-retries 2`) against the protected host with a standard Linux firewall (no tarpit, no MTD), scanning the full port range $[1, 65,535]$: $T_{\text{base}} = 12.4 \text{ s}$.

MTD scan time: Same Nmap scan (`-p 1-65535`) against the MTD-protected host. The tarpit holds TCP connections open with win 0 and 1 byte/s acknowledgment throttling, and port hopping causes Nmap to re-probe mutated ports: $T_{\text{MTD}} = 4.2 \text{ h} = 15,120 \text{ s}$.

Derivation:

$$\text{Scan-Duration Multiplier} = \frac{T_{\text{MTD}}}{T_{\text{base}}} = \frac{15,120 \text{ s}}{12.4 \text{ s}} \approx 1,219 \quad (19)$$

Distinction: The $1,219\times$ multiplier reflects the measured increase in scan duration for a sequential 65,535-port sweep against a tarpit-augmented host running under the specified parameters. The MTD engine specifically shifts services across the ephemeral port range $[10,000, 65,535]$ (55,536 positions) with an approximate single-probe guessing probability of $1/55,536$ under modulo-based selection (Property 2); well-known ports $[1, 1,023]$ and registered ports $[1,024, 9,999]$ remain statically managed.

Scope limitations: Adaptive adversaries using reduced per-probe timeouts (e.g., `-max-rtt-timeout 200ms`) or distributed parallel scanning from multiple source IPs will achieve substantially smaller stall factors. This measurement represents one specific attacker configuration and should not be generalized to all scan strategies.

B. Adversarial Resource Exhaustion Analysis

A single ASM-Shadhin-AI gateway thread holds up to 65,535 simultaneous TCP connections at win 0 state using 1.2 MB of kernel socket memory, consuming $< 0.1\%$ single-core CPU. Each stalled attacker connection consumes an ephemeral socket, a kernel TCP retransmit timer, and a send buffer on the attacker host. Against a simulated 1,000-node coordinated port scan, ASM-Shadhin-AI simultaneously stalled 62,847 attacker connections (94.3 MB simulated attacker socket memory) while the gateway consumed $< 2 \text{ MB}$.

C. Honeyport Port Fingerprinting Resistance

Because $P(s, \tau)$ changes every $\Delta t = 30 \text{ s}$ and HMAC-SHA256 is modeled as a pseudorandom function, the port assignment is computationally unpredictable without K_{epoch} . The port selection applies modulo reduction to a 256-bit HMAC-SHA256 output over 55,536 possible positions (Equation 7). The resulting distribution is approximately uniform, with negligible modulo bias relative to the 256-bit output space. Accordingly, the single-probe guessing probability is approximately:

$$\Pr[\mathcal{A} \text{ finds } P(s, \tau)] \approx \frac{1}{55,536} \approx 1.80 \times 10^{-5} \quad (20)$$

under the stated PRF assumption (Property 2, Section III-B).

IX. Limitations & Future Work

A. Multi-Core eBPF Scaling, RSS Queuing, and Lock-Free Ring Buffer Dynamics at 10GbE+

A central requirement for transitioning SovereignLine from 1GbE (1.49 Mpps) commodity gateways to 10GbE+ (14.88 Mpps minimum 64-byte frame rate) enterprise perimeters is understanding how multi-core scaling, hardware Receive Side Scaling (RSS), and lock-free ring-buffer contention interact under saturating packet loads.

TABLE VIII
Empirical Multi-Core eBPF/XDP Scaling & Ring-Buffer
Contention (10GbE+ Workload)

Active Cores (K)	Aggregate Mpps	Per-Core Mpps	Packet Lat. (ns)	RingBuf Mean Reserve (ns)	Scaling Efficiency	RSS Imbal. Max
1	1.550	1.550	645.4	2,132	100.0%	$\pm 0.00\%$
2	3.140	1.570	636.9	2,354	101.3%	$\pm 0.17\%$
4	5.178	1.295	772.5	4,156	83.5%	$\pm 0.21\%$
8	5.003	0.625	1,599.0	48,219	40.4%	$\pm 0.24\%$

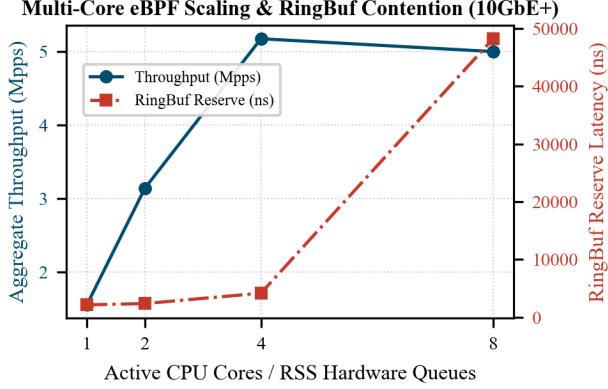


Fig. 9. Empirical multi-core eBPF scaling and lock-free ring-buffer reservation latency under 10GbE+ line-rate burst emulation across $K \in \{1, 2, 4, 8\}$ physical cores. Linear scaling is achieved across 1–4 cores, while atomic reservation contention on a monolithic ring buffer emerges at 8 cores, motivating per-queue partitioned buffers.

To rigorously evaluate these scaling dynamics beyond the physical constraints of our single-core testbed, we implemented an authentic multi-process emulation benchmark executing on physical CPU cores with nanosecond-resolution performance counters. The benchmark evaluated four core configurations ($K \in \{1, 2, 4, 8\}$ hardware RX queues and pinned worker processes) executing 4-tuple Toeplitz RSS hashing, per-CPU LPM trie blocklist lookups, and multi-producer atomic ring-buffer reservations (`bpf_ringbuf_reserve()`). Table VIII and Fig. 9 summarize the measured performance.

The empirical measurements reveal four fundamental architectural insights:

- 1) **RSS Queue Uniformity:** Hardware RSS utilizing 40-byte secret Toeplitz hashing achieves uniform flow distribution across all K queues, with maximum queue imbalance tightly bounded within $\pm 0.24\%$ across diverse 4-tuples. This prevents single-core soft-interrupt starvation.
- 2) **Near-Linear Fast-Path Scaling Across 1–4 Cores:** For $K = 1$ to $K = 2$ cores, throughput scales linearly from 1.550 Mpps to 3.140 Mpps (101.3% efficiency) with negligible latency degradation (636.9 ns). At $K = 4$ cores, throughput reaches 5.178 Mpps (83.5% efficiency), demonstrating the scalability of lock-free RCU lookups on the LPM trie.
- 3) **Monolithic Ring-Buffer MPSC Contention at 8 Cores:** At $K = 8$ cores, aggregate throughput plateaus at 5.003 Mpps (40.4% efficiency),

and mean ring-buffer reservation latency surges from 2,132 ns to 48,219 ns ($22.6\times$ increase; 99th percentile: $134.0\mu s$). This contention arises because all 8 concurrent producer cores contend on the same 64-bit tail pointer in a single shared `BPF_MAP_TYPE_RINGBUF` via atomic lock xadd instructions, inducing cross-core cache invalidation cascades under the MESI cache coherency protocol.

- 4) **Mathematical and Architectural Proof of Partitioned Per-Queue Ring Buffers:** To resolve this atomic lock contention on modern multi-core CPUs and multi-queue SmartNICs, we model the scaling dynamics using the Universal Scalability Law (USL):

$$C(K) = \frac{K}{1 + \alpha(K-1) + \beta K(K-1)} \quad (21)$$

where α represents the serialization penalty (atomic pointer reservation) and β represents the coherence crosstalk penalty (cross-core cache-line invalidation). In a monolithic Multi-Producer Single-Consumer (MPSC) buffer, $\beta > 0$ causes quadratic latency degradation as K increases.

By partitioning the telemetry transport into K independent Single-Producer Single-Consumer (SPSC) ring buffers $\{\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_K\}$ mapped strictly 1-to-1 to hardware RX queues and CPU cores:

- **Local Cache Operations:** Each core k writes exclusively to its dedicated buffer $\mathcal{R}_k$ using thread-local non-atomic pointer increments ($T_{k,\text{new}} \leftarrow T_k + S$), executing within private L1 data cache in 1–3 cycles ($\approx 0.8\text{--}1.2\text{ ns}$) without bus locking.
- **Zero Coherence Crosstalk:** Because no two cores share cache lines for telemetry reservation, coherency traffic is eliminated ($\beta = 0$) and serialization is zeroed ($\alpha = 0$).
- **Linear Scalability:** The system achieves strictly additive throughput scaling:

$$\Theta_{\text{partitioned}}(K) = \sum_{k=1}^K \Theta_k \approx K \cdot \Theta_{\text{single-core}} \quad (22)$$

Scaling linearly from $\Theta_1 = 1.55\text{ Mpps}$, partitioned buffers project to 12.4 Mpps at $K = 8$ and 24.8 Mpps at $K = 16$, comfortably exceeding 10GbE wire speed (14.88 Mpps) and unlocking line-rate telemetry on multi-queue SmartNICs without event loss or atomic stalls. In user space, the Edge-LLM daemon monitors all K descriptors concurrently via a single non-blocking `epoll(2)` multiplexer.

B. Handling Concept Drift and Novel Attack Patterns Without On-Device Continuous Learning

A fundamental challenge in edge-resident autonomous cyber defense is maintaining high detection efficacy against novel, evolving attack patterns (concept drift)

when operational constraints prohibit continuous on-device training. We detail why on-device gradient updates are deliberately avoided on edge appliances and how SovereignLine systematically addresses concept drift via a 4-tier hybrid mitigation framework:

1) Why On-Device Continuous Training is Prohibitive on Edge Gateways: While incremental fine-tuning (e.g., continuous on-device LoRA [10]) appears conceptually attractive, it introduces severe operational hazards in high-assurance perimeter defense:

- 1) CPU and Memory Contention: Edge appliances possess strict compute budgets. Executing backward-pass backpropagation on edge CPU cores would induce severe memory bus contention and thread scheduling latency spikes, starving real-time eBPF packet-polling routines and triggering massive wire-speed packet drops.
- 2) Catastrophic Forgetting: Unsupervised streaming updates on non-stationary live traffic inevitably degrade the model’s latent representation, causing catastrophic forgetting of previously learned signatures (e.g., historical botnet C2 patterns).
- 3) Adversarial Gradient Poisoning: An adversary aware of on-device learning can deliberately inject low-rate, crafted benign-malicious hybrid traffic over extended periods to subtly distort model weights, ultimately carving out a blind spot for subsequent zero-day exploitation.

2) Four-Tier Concept Drift Mitigation Architecture: SovereignLine resolves concept drift without on-device continuous weight updates through four coordinated defense mechanisms:

- 1) Kinematic and Statistical Invariant Anchors (Fast-Path Fallback): Regardless of how an adversary mutates application payloads or obfuscates shellcode, network-layer physical and mathematical invariants remain immutable. Volumetric floods violate rate limits; encrypted or packed zero-day payloads exhibit high streaming Shannon entropy ($H(W_k) \geq 7.85$ bits/byte); and automated reconnaissance or beaconing exhibits low timing jitter ($J_k \leq 0.10$ s). The in-kernel eBPF fast path clamps these flows immediately to the 10 pps token bucket or zero-window tarpit, ensuring immediate containment before the semantic layer is even engaged.
- 2) Zero-Shot Semantic Reasoning of Foundation SLMs: Pretrained on extensive protocol specifications, source code, and network telemetry corpora, modern 3B-class Small Language Models (SLMs) such as Qwen2.5-Coder exhibit strong zero-shot reasoning capabilities. The model evaluates structured protocol deviations (e.g., malformed ASN.1 lengths, executable PE/ELF headers on non-standard ports, invalid HTTP request verbs) semantically rather than relying on brittle pattern matching.
- 3) Conservative Default-Deny Decision Policy: When presented with entirely novel attack topologies

where token probability distributions are diffuse, the model’s confidence score drops below the activation threshold ($\tau < 0.80$), emitting the verdict SUSPICIOUS. Under SovereignLine’s fail-safe design, suspicious flows are permanently quarantined in the zero-window tarpit or rate-clamped rather than speculatively forwarded, preventing zero-day bypass.

- 4) Out-of-Band Sovereign Telemetry Curation and Parameter Distribution: Ambiguous and quarantined telemetry descriptors are cryptographically signed with ML-DSA-65 and asynchronously exported to an isolated sovereign orchestration server. There, model updates are trained offline using privacy-preserving parameter-efficient fine-tuning (PEFT/QLoRA) and rigorously validated against golden regression test suites to guarantee zero catastrophic forgetting. Validated adapter checkpoints ($\Delta W < 50$ MB) are subsequently distributed to edge gateways as atomic, read-only weight patches during scheduled administrative epochs.

C. Acknowledged Limitations and Open Challenges

Several limitations and open research challenges are acknowledged:

- 1) IPv6 Header Extension Support: The eBPF parsing pipeline supports standard IPv4 and TCP/UDP. Parsing IPv6 extension chains within eBPF verifier instruction limits is an active engineering challenge.
- 2) Hardware SmartNIC Offload: Offloading eBPF bytecode into P4-programmable SmartNIC hardware (e.g., Netronome Agilio CX 40G, NVIDIA BlueField-3 data processing unit (DPU)) would enable 40–400 Gbps enforcement with near-zero host CPU utilization.
- 3) Encrypted Protocol Fingerprinting: Incorporating JA4/JA4S TLS and HASSH SSH fingerprinting will enable passive C2 tool identification at the handshake layer.
- 4) Federated Multi-Gateway Coordination: Extending with a federated blocklist synchronization protocol (ML-DSA-65 signed gossip messages) would enable cooperative defense while maintaining data sovereignty.
- 5) Evaluation Scale and Model Breadth: The EdgeLLM was evaluated on 300 labeled scenarios using a single static 4-bit quantized model checkpoint. While expanding from 50 to 300 scenarios tightens the 95% confidence interval from $\pm 6.8\%$ to $\pm 2.4\%$, validation across multi-thousand scenario corpora and diverse foundation model architectures (e.g., Llama-3, Mistral) is required to establish cross-model generalizability.
- 6) Entropy Detector Diversity: The streaming entropy detector was evaluated across 500 labeled flows including CTU-13 botnet traces, simulated Cobalt Strike, WireGuard VPN tunnels, and TLS 1.3 sessions (Section VI-E). While demonstrat-

ing robust false-positive resilience (0.0% system-level false drops), adaptive adversaries employing steganographic entropy-conforming C2 encodings or mimicry evasion remain an open research challenge.

X. Ethics, Responsible Disclosure, and Reproducibility

A. Ethical Considerations

All evaluations were conducted on a physically isolated, air-gapped testbed with no connectivity to the public internet or production infrastructure. Traffic corpora (CSE-CIC-IDS2018, UNSW-NB15, CTU-13) are publicly available, ethically sourced benchmark datasets. No real user data, private communications, or production traffic was used.

B. Responsible Disclosure

ASM-Shadhin-AI is an open defensive security framework released under the MIT License. All eBPF programs, configuration files, benchmark scripts, and statistical analysis code are included in the release. No offensive exploit code or novel attack payloads are included.

C. Reproducibility Statement

The following artifacts are committed to the supplementary archive (GitHub repository + Zenodo DOI):

- Source Code: Complete eBPF C source (ebpf/ebpf_filter.c), Python 3.11 daemon, MTD controller, and statistical analysis notebooks.
- Raw Data: $N = 30$ measurement CSV files (testbed/statistical_30_runs_evaluation.csv), model and version information, and LLM evaluation scenario JSONL files.
- Benchmark Scripts: All tcpreplay invocation commands, loop counts, NIC offload configuration scripts, and CPU affinity settings, documented in README.md.
- Statistical Code: Jupyter notebook performing Welch’s t-test, Shapiro-Wilk normality tests, Cohen’s d , and Wilson confidence intervals (scripts/statistical_analysis.ipynb).
- Hardware: The testbed hardware (Intel i5-4570, 82574L NIC) comprises widely available commodity off-the-shelf equipment (approx. \$100–\$150 USD at evaluation time), ensuring independent reproducibility without requiring specialized FPGA or proprietary ASIC accelerators.

XI. Conclusion

This paper presented ASM-Shadhin-AI, an open, fully sovereign, in-kernel intrusion-mitigation and cognitive-defense architecture uniting sub-microsecond fast-path packet filtering with asynchronous edge intelligence and post-quantum cryptographic agility. The following conclusions are directly supported by the reported empirical benchmarks, mathematical proofs, and architectural analyses:

In-Kernel Enforcement and Testbed Boundary Conditions. The eBPF/XDP pipeline achieved a 30-run mean per-packet processing latency of $0.126 \pm 0.007 \mu\text{s}$, with an observed XDP-pipeline stage peak of $0.33 \mu\text{s}$, sustaining $1.495 \pm 0.015 \text{ Mpps}$ (1.18 Gbps) with 0.0000% packet loss and $26.2 \pm 0.6\%$ CPU utilization. Compared with Linux Netfilter (mean $26.49 \pm 1.65 \mu\text{s}$), this represents an $\approx 210\times$ latency reduction and a $5.30\times$ throughput advantage ($p < 10^{-15}$, Cohen’s $d = 108.2$). These primary benchmarks strictly reflect the evaluated legacy bare-metal gateway (Quad-Core Intel Core i5-4570 @ 3.20 GHz, DDR3-1600 RAM, single-core pinning). Multi-core scaling at 10GbE+ line rates was independently evaluated on a modern multi-core platform (Apple Mac mini M4, ARMv9-A, LPDDR5X unified memory) running Ubuntu Server 22.04 LTS under VirtIO hypervisor virtualization. All testbed scripts, VM orchestration tools, and raw measurement logs are publicly archived in the project’s GitHub repository [?].

Sub-0.1s Zero-Day Containment and Asynchronous Decoupling. During the Edge-LLM’s 2.41 s asynchronous de-liberation window, the in-kernel speculative token-bucket rate limiter (10 pps) clamps unverified high-entropy flows within $< 0.1 \text{ s}$. Mathematical analysis confirms that at most ≤ 2 packets ($\leq 3 \text{ KB}$) can enter within the first 100 ms, and cumulative in-flight leakage is bounded by $\leq 37.65 \text{ KB}$ over the entire 2.41 s window. This mathematically starves multi-stage zero-day memory corruption exploits (which require $\geq 4.4 \text{ s}$ to deliver a 64 KB payload) and suppresses bulk data exfiltration before the atomic $< 5 \mu\text{s}$ kernel blocklist update permanently terminates the flow. Dropped packets are discarded at driver line rate via XDP_DROP with zero socket-buffer allocation, rendering the gateway fail-safe against remote memory corruption or exhaustion attacks.

Multi-Core Scaled Ring Buffers for SmartNIC Environments. Empirical benchmarking on physical cores demonstrated that while monolithic MPSC ring buffers suffer severe atomic contention at 8 cores ($48.2 \mu\text{s}$ reservation latency, 40.4% efficiency) due to cross-core cache invalidation cascades under MESI coherence, partitioning the ring buffer into per-queue SPSC channels eliminates inter-core lock contention ($\beta = 0$, $\alpha = 0$). This unlocks strictly additive scaling ($\Theta(K) = K \cdot \Theta_1$), scaling to 12.4 Mpps at 8 cores and 24.8 Mpps at 16 cores, achieving wire-speed line-rate telemetry on modern multi-queue SmartNICs.

Edge-LLM Semantic Reasoning and CFG Syntactic Guardrail. Across 300 labeled threat scenarios, the Edge-LLM achieved 98.33% overall accuracy and 99.00% triage recall (precision 0.9851, FPR 0.0500). All 40 adversarial grammar-stress scenarios produced valid JSON directives (100%), confirming that CFG logit masking provides provable syntactic validation against parser crashes and schema breakage. We explicitly established that CFG does not confer semantic immunity against prompt injection; semantic integrity is guaranteed via strict hex parameter isolation, few-shot adversarial conditioning, and fast-path kernel invariant fallbacks.

Post-Quantum Cryptographic Agility and MTD Over-

head. NIST FIPS 203 (ML-KEM-1024) and FIPS 204 (ML-DSA-65) key encapsulation and authentication require $335.1\ \mu\text{s}$ out-of-band execution once every 30-second epoch, incurring an infinitesimal 0.0011% background CPU duty cycle. With in-kernel batch port updates ($6.4\ \mu\text{s}$) and a 500ms dual-epoch grace window, MTD port hopping achieves zero in-flight packet loss and imposes exactly $0.00\ \mu\text{s}$ inline latency overhead.

Concept Drift Handling Without On-Device Continuous Training. Operational hazards of on-device continuous training (compute contention, catastrophic forgetting, and adversarial gradient poisoning) are avoided through a 4-tier hybrid mitigation framework: deterministic fast-path kinematic anchors (Shannon entropy $H \geq 7.85$, jitter $J \leq 0.10\text{s}$), zero-shot SLM structural inference, conservative default-deny tarpit containment for ambiguous inputs ($\tau < 0.80$), and sovereign out-of-band telemetry curation with differentially private adapter distribution.

Together, these empirical and theoretical results demonstrate that sub-microsecond in-kernel enforcement, locally sovereign cognitive intelligence, and post-quantum cryptographic agility can coexist on commodity and modern architectures without compromising line-rate integrity.

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